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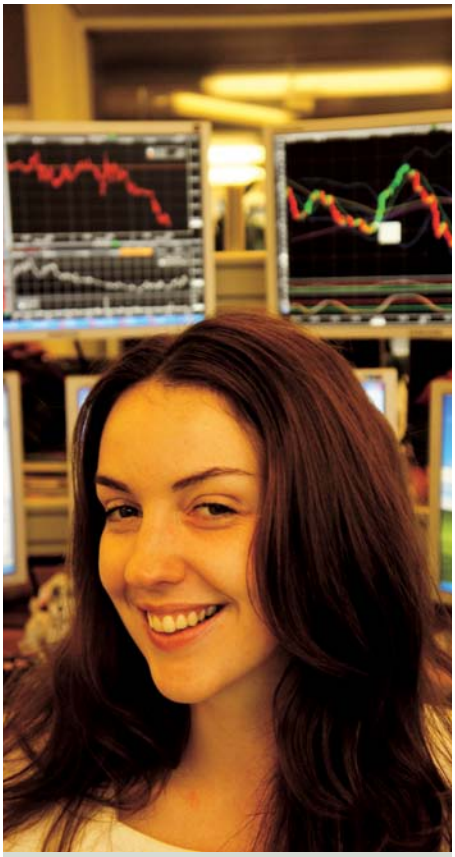
INSIGHT SPECIAL: COFFEE FUTURES 101

CBS&A Coffee Business Services & Academy, a Volcafe Initiative



__Kirsty MacGregor__ is Volcafe's Head of Research. She is an industry leader in the field of research, insights and forecasting, plus a key part of the CBS&A team of coffee experts. In this __Insight Special__ we explain the basics of coffee futures.

HIGHLIGHTS

All commodity markets tend to be volatile. Agricultural commodity markets, with their vulnerability to weather shocks, are particularly volatile.

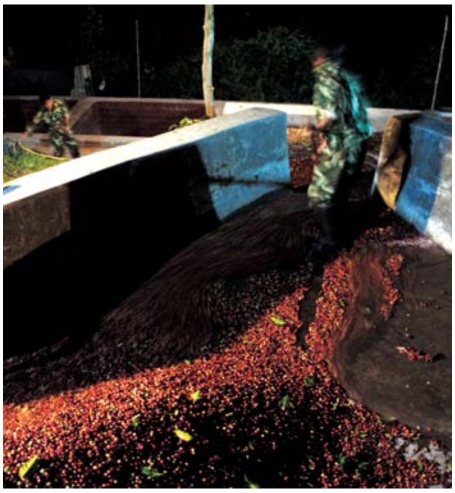
The coffee market especially has the well-deserved reputation of boom and bust in relation to price levels. The act of bringing coffee from seedling to cup is a journey set against a background of high price uncertainty, which could paralyse market participants and prevent them extending business beyond just the near horizon.

This is where a risk management tool, developed over many years, steps in; the exchange-traded coffee future contract.

THE BASICS

A coffee future is a standardised contract that trades on a commodity exchange, which prices coffee for many months in advance, and can be settled by physical delivery of a specified type of coffee at a certain time. When you buy or sell a coffee future, you are agreeing to the obligation to buy or sell the physical underlying coffee at a certain price at a certain point in the future. However, coffee futures are usually not settled with physical delivery, unless market conditions recommend it.

The majority of coffee futures users are hedgers. Hedging is the practice of reducing the price risk of a position in physical coffee, by taking the opposite position in the futures market. Physical market losses would result in future market gains, and physical market gains would result in futures markets losses.



A hedger's futures market position is normally liquidated on the futures market itself, once the physical position (for which the futures hedge was done) has passed.

MAPPING THE COFFEE FUTURES MARKETS

There are three main futures exchanges for coffee, with three different futures contracts which go some way to covering the diversity of global coffees:

- the ICE exchange in New York trades the 'C' contract where the underlying coffee is **washed arabica**
- the Liffe exchange in London trades a contract with **robusta** as the underlying
- the BM&F exchange in Sao Paulo trades a **natural Brazilian arabica** futures contract.

Along with having different physical coffee as the underlying, these three futures contracts have different standardised pricing units, lot sizes, trading hours, delivery months, etc.

THREE MAIN FUTURES EXCHANGES FOR COFFEE AT A GLANCE

	ICE 'C' CONTRACT	LIFFE ROBUSTA CONTRACT	BM&F ARABICA CONTRACT
Where	New York	London	Sao Paulo
Trading Hours (GMT)	08.30 - 19.00	09.00 - 17.30	11.00 - 18.45 & 19.30 - 20.00
Size of one lot	37,500 lbs / 17 tonnes	10 tonnes	100 bags of 60 kg / 6 tonnes
Price Quotation / Unit	US cts/lb	USD/tonne	USD/60kg
Delivery Months	Mar, May, July, Sep, Dec	Jan, Mar, May, July, Sep, Nov	Mar, May, July, Sep, Dec
Physical Settlement	Certain quality of washed arabica coffee from 19 specified countries, Brazil will only be included from 2013	Certain quality of robusta coffee from any country	Certain quality of Brazilian arabica natural coffee

WHY FUTURES?

Of course, any coffee market participants can enter into forward contracts directly with each other, not through the terminal market / exchange. However the coffee futures contract has the benefit that the exchange is always the financial counterparty to the futures market participants when a bid and offer (coming from market participants) are matched / filled. This reduces counterparty risk. Coffee future trading also has the advantage of mass access and plenty of participants willing to take the opposite side of a trade. The three main coffee futures contracts which are traded in New York, London and Sao Paulo, can be bought and sold by anyone with an account. The coffee futures markets allow mass participation, liquidity, and thus, in theory, efficient price discovery.

WHO USES FUTURES MARKETS:

The hedgers – those who grow, buy, sell and process physical coffee

Sometimes it is preferable for a farmer and a roaster to plan a price (or 'lock in' a price) at which they will sell or buy their raw product in the future.

PRODUCERS – THE NATURAL 'LONGS' OF THE COFFEE WORLD

Farmers know that, barring natural disasters, they will have coffee to sell once the harvest comes in. While they may not want to, or be able to, sell to their normal buyers today, they can sell to the futures market. This means they can avoid the risk of the market falling by the time the harvest is in and they are ready to sell the actual crop.

December 2011

Sells one lot coffee future at
230cts/lb
for March 2012 delivery

Either / Or

by March 2012

Market Falls to 200cts/lb
- Buys back Future at 200cts/lb (pockets 30cts/lb)
- Sells to trade buyer at 200cts/lb (loses 30cts/lb)

NET Price received = 230cts/lb

Market Rises to 260cts/lb
- Buys back Future at 260cts/lb (loses 30cts/lb)
- Sells to trade buyer at 260cts/lb (pockets 30cts/lb)

NET Price received = 230cts/lb

In this example, the farmer liquidates his short futures position by buying back his futures on the exchange and selling his physical coffee locally to the trader. In theory, he could equally well deliver the coffee to a warehouse approved by the exchange, but in practice this rarely happens.

ROASTERS – THE NATURAL 'SHORTS' OF THE COFFEE WORLD

The roaster has a different concern. He is worried that the market will rise by the time he is able to buy the coffee, so he buys from the futures market instead to hedge this risk.

December 2011

Buys one lot coffee future at
230cts/lb
for March 2012 delivery

Either / Or

by March 2012

Market Falls to 200cts/lb
- Sells back Future at 200cts/lb (loses 30cts/lb)
- Buys from trade at 200cts/lb (gains 30cts/lb)

NET Price paid = 230cts/lb

Market Rises to 260cts/lb
- Sells back Future at 260cts/lb (gains 30cts/lb)
- Buys from trade at 260cts/lb (loses 30cts/lb)

NET Price paid = 230cts/lb

This is the theory, but in practice not many producers, and a surprising amount of roasters, do not operate their own futures accounts. Many producers prefer to sell outright on the day of sale with no futures trade previously, and many roasters buy coffee differentially on a 'price-to-be-fixed' basis. When physical coffee is traded price-to-be-fixed, a coffee roaster agrees to purchase coffee at a differential to the futures market, and only later do they decide to 'fix' the final price against the price of a future that the trader has to purchase at the moment/in the manner the roaster chooses. The 'fixation desk' at a tradehouse is a busy place indeed. This means that the roaster's involvement in the futures market can be limited to telling the coffee tradehouse at which time to fix the final price, with the futures only being traded in the tradehouse's account. Of course, larger roasters with futures accounts will be opting for 'price coverage' in addition to this, and those futures they do own, will be transferred to the tradehouse at the time of fixing the physical contract, in a trade called 'AA' – or against actuals.

TRADEHOUSES

Tradehouses like Volcafe are the key link between the usually geographically distanced producers and roasters (for example the producer in Guatemala and the roaster in Japan). Our operational capabilities in key origins allow us to procure, sort and prepare green coffee according to the specific requirements of our customers. Expertise in coffee qualities and sustainability know-how help us to meet and even exceed the expectations of our most discerning customers; and our reputation for reliability and service is extremely important to us. The nature of coffee production and consumption however generates vast price risks for a coffee trader; we use the coffee futures market to reduce this risk to manageable levels.

Usually, a coffee tradehouse will:

- buy from the producer on a outright basis.
- sell to the roaster on a "price-to-be-fixed" basis.

For example:

- March 2011 – a tradehouse sells 17 tonnes of coffee to a roaster at +5 cts/lb over the ICE 'C' contract, price to be fixed, buyers call, November shipment.
- August 2011 – the tradehouse can now buy the coffee from the producer, at 273 cts/lb outright. The tradehouse hedges this outright purchase around this time with the sale of 1 lot of Sep 11 future, at 270 cts/lb. Thus the buying differential is established at +3 cts/lb over the 'C'.
- October 2011 – the roaster phones up the tradehouse to fix the price of their purchase. The tradehouse buys 1 lot of Dec 11 future at 230 cts/lb. This closes the trader's futures trade with a gain of 40 cts/lb (the Sep/Dec futures 'spread is managed elsewhere in their 'book'). The physical coffee sale is now fixed at a price of 235 cts/lb. This closes the physical trade with a loss of 38 cts/lb.

TO SUMMARISE

- The trader sold futures at 270 cts/lb, and bought futures at 230 = gain of 40 cts/lb
- The trader bought physical coffee at 273 cts/lb, and sold at 235 = loss of 38 cts/lb.
- Trader makes 2 cts/lb.

In the trader's eyes, he has made a "differential" gain of 2 cts/lb, since he bought at +3 and sold at +5. The futures transactions were just a mechanism to allow him to manage his risk between purchase and final fixation of sale price. The outright price is much more volatile than the differential. The futures market helps to reduce the risk, and with more willing participants, allows for more efficient price discovery.

WHO USES FUTURES MARKETS:
The funds and speculators

It is not just commercial hedgers who trade coffee futures (those who need to reduce the price risk of physical coffee positions). In recent years, more and more financial investors (a nice way of saying speculators) have sought exposure to commodities as an asset class. In the ICE 'C' contract at least, those classed as commercial hedgers account for around two-thirds of the open interest, while financial investors account for the rest. These non-commercial participants include index funds, macro funds and system funds, a fund's inclusion in one of these groups being marked out by its usual strategy and style of trading.

TERMINAL TERMINOLOGY: THE BASICS

COFFEE FUTURE CONTRACT:
A standardised contract that trades on an exchange, which prices coffee for many specified months in advance, and can be settled by physical delivery of a specified type of coffee at a certain time.

FUTURES MARKET / COMMODITY EXCHANGE / TERMINAL MARKET:
All these terms mean the same thing; a market where futures contracts are traded.

DERIVATIVE: An umbrella term for any contract (standardised and non-standardised) that is traded in or outside of exchanges, which prices an underlying asset through trading the right or obligations to trade that underlying asset at a point generally in the future. Derivatives are for the most part used to hedge risk. A coffee future comes under this umbrella term.

GOING LONG COFFEE FUTURES: Buying coffee futures, which basically means you are agreeing to buy a certain volume of the underlying physical coffee at a certain price at a specified point in the future.

GOING SHORT COFFEE FUTURES: Selling coffee futures, which basically means you are agreeing to sell a certain volume of the underlying physical coffee at a certain price at a specified point in the future.

PHYSICAL UNDERLYING: The actual type of coffee which the futures contract claims to be pricing. The physical delivery settlement of a coffee futures market is key; the price of the future at the time of delivery must reflect the underlying price of the coffee it claims to represent. Otherwise the market participants who use coffee futures as a hedge (see below) for physical business are increasing their risk rather than reducing it. However, coffee futures are usually **not** settled with physical delivery, unless market conditions recommend it.

DELIVERY MONTHS: A coffee future always has a delivery month (also called contract month), by the end of which the trade must be (and is most commonly) liquidated either through the opposite futures trade (see below), or through physical delivery of coffee in exchange-licensed warehouses. These delivery months have codes, for example U12 refers to September 2012, or H13 refers to March 2013. These months are the same calendar months every year, for each specific exchange.

NOTICE PERIOD: The multi-day period of time at the end of the trading life of a delivery month, when any trader with an open position be notified or can notify that the futures he is long or short will be settled with physical delivery.

EXPIRY: When the delivery month contract ceases to trade.

SPOT: Used to indicate immediacy. Employed to describe physical coffee that is available at once (already shipped). Also used in the futures market to refer to a futures delivery month most commonly during its notice period.

1ST MONTH / SPOT MONTH / NEARBY MONTH / FRONT MONTH: Various terms to describe the delivery month closest to expiration.

2ND OR 3RD MONTH ETC / FORWARD MONTH: Terms to describe the delivery months coming after the front delivery month.

LIQUIDATE / CLOSE A POSITION / SQUARE:
The act of cancelling a futures position by carrying out the opposite futures trade to the one already done. A trader who has bought futures can just sell the same number of lots of futures in the same delivery month, before expiry of this delivery month (and most straightforwardly before the notice period), to cancel all obligations.

DELIVERING / TENDERING / ISSUING:
Providing (selling) the physical underlying coffee, in the notice period of the delivery month, in order to settle a short futures position.

STOPPING / RECEIVING / TAKING DELIVERY:
Accepting (buying) the physical underlying coffee, in the notice period of the delivery month, in order to settle a long futures position.

LOT: The unit of one future. The size of one lot varies per exchange. NB futures prices are usually quoted in smaller units than the price of one full lot. Even though the Liffe robusta contract is traded in lots of 10 tonnes, the price is always quoted in US dollars per 1 tonne.

HEDGE: The practice of reducing the price risk of a position in the physical market by taking the opposition position in the futures market. Someone who has bought physical coffee would sell futures, and someone who has sold physical coffee would buy futures. Physical market losses would result in future market gains, and physical market gains would result in futures markets losses.

COFFEE DIFFERENTIAL: Hedging results in a 'differential' or 'basis' position. The difference in price of the physical coffee versus the futures price. It is always calculated physical coffee price less futures price. The price volatility of coffee differentials is the main price risk left after a physical coffee position has been hedged through the coffee futures market.

ORDER: An instruction to buy or sell a certain volume of a certain contract of a certain delivery month at a certain price. Nowadays normally entered through electronic trading systems.

BID: An order to buy.

OFFER: An order to sell.

LIMIT ORDER: An order to sell / buy down or up to a certain price limit.

MARKET ORDER: An order to sell / buy at the nearest bid / offer.

STOP ORDER: An order to buy or sell only once the market reaches a certain level, usually in order to liquidate a losing trade.

GTC: An order that is Good Til Cancelled – ie it stays in the market for any number of days until it trades or is cancelled.

DAY ORDER: An order that automatically cancels if it is not traded / filled.

FILLS: Orders which are successfully matched to a corresponding bid or offer, ie they are traded / executed.

MINIMUM PRICE MOVEMENT:
The smallest increment of price movement allowed per futures contract. In ICE, it is 0.05 US cts/lb, in Liffe 1 USD, in BMF 0.05 USD / 60kg.

POINTS: In the ICE 'C' contract, 0.01 US cts/lb is referred to as points. When the 'C' contract moves by 5 cts/lb, one can say it has moved by 500 points.

FND: First Notice Day, the first day of the notice period.

INITIAL MARGIN: A 'deposit' posted by a trader/tradehouse to the exchange when a trade is opened.

VARIATION MARGIN: The value of the position marked at current prices.

MARGIN CALL: A back-and-forth between the exchange and the trader to increase or decrease their margin deposit with the clearing house, either due to a market move, a change in position or exchange rules.

CLEARING: The process to accept and record trades and settle margins on a daily basis.

CLEARING MEMBER: All traders must trade through a clearing member of an exchange, usually a bank or a brokerage firm.

BULLISH: A market outlook where you think the price will go up.

BEARISH: A market outlook where you think the price will go down.

OUTRIGHT POSITION: Generally refers to a stand-alone position either in the futures or physical market, that is not offset by another trade.

BOOK: A summary of all the trades and offsetting trades that a trader or a company takes.

SPREAD/SWITCH: Terms which can describe the price difference between delivery months of the same futures contract. Spread can also refer to the price difference between two similar markets, like arbitrage.

ROLLING: If a trader (for example a hedger or a speculator with a long-term trade outlook) wants to maintain his futures positions past the expiry date of the delivery month in which he has a position, he can 'roll' it to the next delivery month. For example, if a trader has bought January 2012 futures, he rolls this position to March 2012 by simply selling the January contract and buying the March contract. This particular 'roll' trade of a long position can be called 'selling the spread'.

ARBITRAGE: The price difference between two similar markets. In coffee futures, the New York / London 'arb' is classically quoted New York less London, most of the time creating a positive number, while the New York / BMF arb is classically quoted BMF less New York, mostly creating a negative number, as an easy comparison to the physical differential of Brazilian coffee against the New York market.

VOLUME VS OPEN INTEREST: Volume is the number of lots traded (ie a matched bid and offer is one lot traded) in a specified time period, while open interest is the total number of unsettled (not liquidated or expired) lots at a particular point in time.

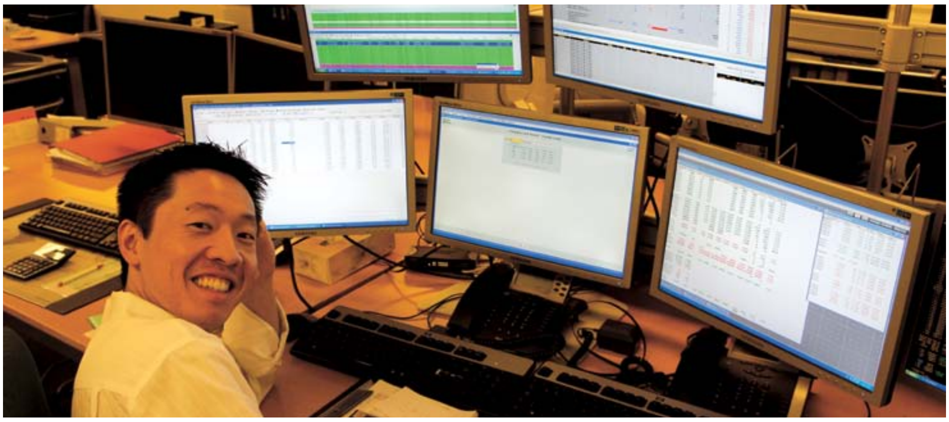
AAS: An abbreviation of against actuals, a simple exchange between two hedgers who want to exchange futures in reference to a physical trade. The futures are traded through the exchange, but flagged to be only traded by the agreed two hedgers.

FUTURES PIT / FLOOR: The (now non-existent) places where coffee futures were traded between floor brokers in the open outcry system. Market participants would have to phone their floor broker to place an order. Only coffee options in New York are still traded in the pit. Coffee futures are now traded electronically through various software platforms, directly by hedgers and speculators.

MARKET STRUCTURE / FORWARD CURVE: Both terms describe the various price levels of delivery months in relation to each other.

CONTANGO (CARRY) MARKET STRUCTURE: When the 1st delivery month is a lower price than the 2nd delivery month. This implies sufficient spot or nearby supply. It is also called a carry market, because this market structure supports the carry trade. That is, if a trader owns fully hedged physical coffee, the short hedge (short futures) in the front month can be bought back, then this futures is 'rolled' or sold again higher in the next month, thus creating a 'profit' that covers the cost of carrying that coffee in the next couple of months: ie financing, insurance and storage rent.

BACKWARDATED (INVERSE) MARKET STRUCTURE: The opposite of a contango market, where the front month is more expensive than the next month, implying a lack of spot or nearby supply in a well-regulated market. A backwardated or inverse market will give the incentive for a coffee owner to sell that coffee, as it causes a 'loss' now to roll the hedge (the trader has to buy the 1st month at a more expensive price than the sale of a future of the 2nd month).



Moreover the physical coffee can be priced for sale against the more expensive front month.

EXCHANGE-CERTIFIED STOCKS: Coffee stocks that are certified by the exchange as suitable for delivery to settle a futures contract.

TENDERABLE PARITY: the price level of physical coffee offered in a producing country which corresponds to a break-even for a trader who would buy that coffee, and then tender it to the exchange as physical settlement of a short futures position taken as an immediate hedge when buying the physical coffee. A coffee differential has reached tenderable parity when the (minus) differential is the same as the total cost of transport, finance, insurance and certification of that coffee to the exchange.

DELIVERY PORTS: The ports specified in the coffee futures contract, where exchange-licensed warehouses are located. It is in these warehouses that the exchange-certified coffee stocks are stored, and where these coffee stocks can be delivered or received as settlement of an expiring coffee futures contract.

TECHNICAL / CHART ANALYSIS: Used by traders to try to predict price direction, based on past volume and price action.

CANDLESTICKS: Appearing on the chart looking like candles, these show the open /high / low / close of the daily / weekly / monthly etc price moves of coffee, with different shades whether the close was below or above the opening price.

POSITION LIMIT: The number of lots a trader can be long or short in a certain delivery month or all months, regulated and enforced by the exchange. The ICE 'C' contract and the BMF arabica contract have position limits, the Liffe robusta contract does not.

LIQUID MARKET: A market with enough participants and a large enough volume of bids and offers to allow for large transactions to be filled without a price spike or steep price drop. Opposite of thin market.

CFTC: A regulatory body created by the US congress in 1974, the Commodity Futures Trading Commission regulates commodity futures and options markets in the US.

COMMITMENT OF TRADERS: This report is published weekly by the CFTC. It classifies traders into separate commercial and non-commerical categories, and publishes the net position per category, allowing for some market transparency for the New York ICE 'C' contract.